

Regime 2: Bear Trend

Q1

EURUSD · 60-minute bars (H1) · n=25,000 bars

Symbol: EURUSD

Bar timeframe: 60 minutes (H1)

Total bars in backtest sample: 25,000

Bars with this regime (primary label): 100 (0.40% of sample)

Quadrant: Q1

Win rates (wr_1r-wr_4r): fraction of trades where price reached $\geq 1R \dots \geq 4R$ favorable excursion before ATR-based stop (see scorecard script).

rank_score: blends wr_1r-wr_4r and lightly penalizes strategies with trades < 8 .

Next pages: Regime52Params (classifier + indicator windows), this regime's taxonomy line, then each strategy's name, signal_kind, side rule, and numeric thresholds.

Regime detection — parameter terms and values (Regime52Params)

EURUSD · 60-minute bars (H1) · n=25,000 bars

Regime52Params — shared by the 52-regime classifier and signal indicators:

adx_period:	14
rsi_period:	14
ema_slow:	50
ema_ob:	20
ema_fast:	12
structure_lb:	10
mom_lookback:	60
bb_period:	20
z_period:	20
squeeze_win:	250
vol_cluster_win:	48
hmm_vol_quantile:	0.75

Classifier rules (thresholds per regime id, claim order, optional ctx_* columns) live in `forex_regime/regimes52/classify.py` — add context columns to activate macro/event regimes when those rows have data.

This regime (id 2) — taxonomy terms
EURUSD · 60-minute bars (H1) · n=25,000 bars

Section: SECTION 1 — Trend

Narrative / rule summary:
Bear Trend — ADX above 40, RSI below 45, LH/LL structure, price below 50 EMA

Strategies for regime 2 — names, keys, sides, signal values

EURUSD · 60-minute bars (H1) · n=25,000 bars

Sample: EURUSD · 60-minute bars (H1) · n=25,000 bars

Strategy rows for this regime (terms). Entries are counted only when regime52_id matches this regime; signal math is shared across all regimes in the same quadrant— only labels differ.

- R02-S01: TSMOM / vol-managed pullback — Bear Trend
signal_kind='pullback_with_trend'; side=+1 → only bullish rule arm fires long
- R02-S02: Channel / Donchian breakout — Bear Trend
signal_kind='breakout_with_trend'; side=+1 → only bullish rule arm fires long
- R02-S03: Cross-sectional momentum add-on — Bear Trend
signal_kind='continuation_bar'; side=+1 → only bullish rule arm fires long
- R02-S04: Low-vol trend drift (defensive gross-up) — Bear Trend
signal_kind='trend_drift'; side=+1 → only bullish rule arm fires long

Signal kinds — numeric values and windows (same for every regime using that kind):

[pullback_with_trend]

EMA windows (Regime52Params): ema_fast=12, ema_ob=20, ema_slow=50
Long: $\text{low} \leq \text{ema_ob} \ \& \ \text{close} > \text{ema_ob} \ \& \ \text{close} > \text{ema50} \ \& \ \text{ema_fast} > \text{ema50}$
Short: $\text{high} \geq \text{ema_ob} \ \& \ \text{close} < \text{ema_ob} \ \& \ \text{close} < \text{ema50} \ \& \ \text{ema_fast} < \text{ema50}$

[breakout_with_trend]

Donchian high/low: rolling $\max(\text{high}) / \min(\text{low})$, period=20
EMA filter: close vs ema_slow (period=50)
Long: $\text{close} > \text{prior Donchian high} \ \& \ \text{close} > \text{ema_slow}$
Short: $\text{close} < \text{prior Donchian low} \ \& \ \text{close} < \text{ema_slow}$

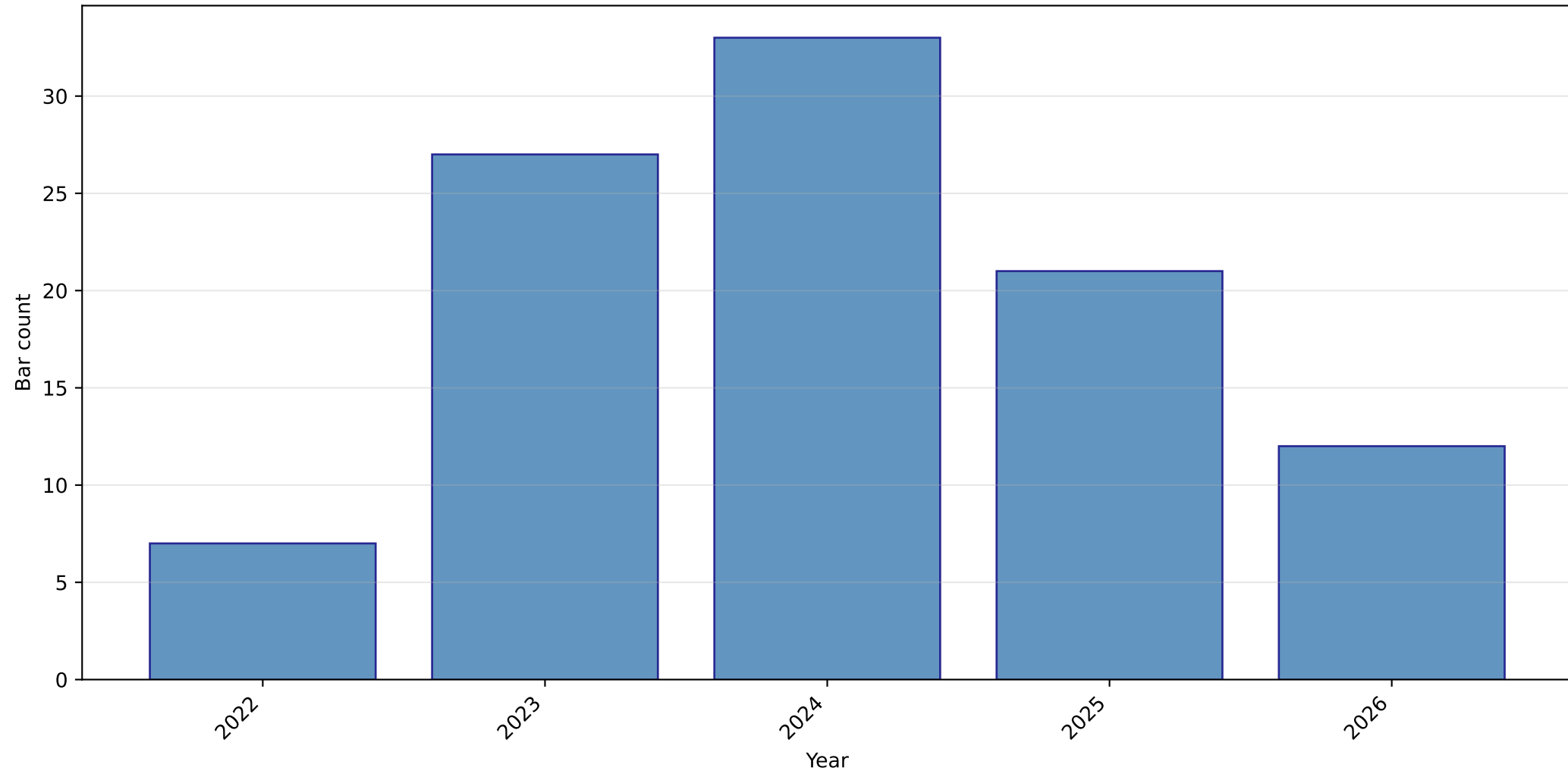
[continuation_bar]

Body filter: $|\text{close} - \text{open}| > 0.45 \times (\text{high} - \text{low})$
EMA trend filter: ema_fast=12, ema_slow=50
Long: bullish bar, body filter, $\text{close} > \text{ema_fast}$, $\text{ema_fast} > \text{ema_slow}$
Short: bearish bar, body filter, $\text{close} < \text{ema_fast}$, $\text{ema_fast} < \text{ema_slow}$

[trend_drift]

ADX filter: Wilder ADX period=14, threshold > 18
EMA slow=50: long if $\text{close} > \text{ema}$ & 1-bar return > 0 & ADX OK; short mirrored

Regime 2 — bar counts by year (UTC)
EURUSD · 60-minute bars (H1) · n=25,000 bars



Regime 2 — bar counts by month (UTC, last 29 months)
EURUSD · 60-minute bars (H1) · n=25,000 bars

